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well upset experimental conditions. Similar data for the nulls of the patterns show equally good agreement between theory and experiment as for the positions of minor lobes.

The relative height of the first minor lobe obtained from experiment as a function of length is compared to the theoretical curve in Fig. 9. Here there is little or no agreement except for order of magnitude; the theory indicates increasing minor lobe height, while experiment leads to decreasing height. The heights of minor lobes are likely greatly affected by the approximations of the theory and by the experimental conditions, whereas the positions of lobes and nulls are less affected by these factors. Thus the good agreement of positions of minor lobes and nulls is not accompanied by good agreement in minor lobe height.

A comparable set of data was compiled for the H-plane patterns. The same agreements and deviations were noted as discussed above for the E-plane, without producing any fundamental differences between the results for the two planes.

SUMMARY

A theory is developed for computing radiation patterns for a dielectric rod used as a termination for a rectangular wave guide. Subject to some restrictions, the theory is applied to the $TE_{0,1}$ mode of excitation in the wave guide, and specific expressions are worked out for the radiation patterns. The expressions contain as parameters the length, width, and thickness of the dielectric rod, and its apparent index of refraction. Suitable choice of the index of refraction produces computed patterns agreeing in considerable detail with patterns for polystyrene rods whose lengths vary from 3λ to 6λ, and whose average thicknesses and widths are the same. Also, a comparable computed pattern is obtained for a considerably thicker rod in the same length range.

It is concluded that, given the physical parameters, the analysis may be used to compute radiation patterns for the $TE_{0,1}$ excitation of rectangular dielectric rods.

The Attenuation of Spherical Shock Waves in Air*

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(Received February 13, 1948)

The peak pressure of spherical shock waves from small explosive charges has been determined as a function of distance by measuring the velocity of propagation and applying the velocity-pressure relation derived from the Rankine-Hugoniot equations. The charges were composed of TNT or pentolite (TNT/PETN 50/50 by weight), weighed from 1.45 to 8 lb., and were spherical or cylindrical in shape, except for one series of small rectangular blocks. Measurements on the non-spherical charges were made in the plane through the center of the charge, perpendicular to the axis.

The pressure-distance relations for the four principle charge types are given by the following fitted equations, in which Π represents excess peak pressure in atmospheres, and the distance, scaled according to charge weight, is given by the non-dimensional variable $Z=R/(\rho\tau)^{\frac{1}{3}}$ where R is the distance from a charge of volume τ and specific gravity ρ:

½-lb. rectangular blocks, TNT,

$$\Pi_Q=13.50/Z-769.9/Z^2+36280/Z^3;$$

8-lb. cylinders, Pentolite,

$$\Pi_A=10.49/Z-135.5/Z^2+21070/Z^3;$$

4-lb. cylinders, TNT,

$$\Pi_B=11.34/Z-185.9/Z^2+19210/Z^3;$$

3.75-lb. spheres, Pentolite,

$$\Pi_C=8.63/Z+295.1/Z^2+7823/Z^3.$$

These equations are valid for values of Z between approximately 18 and 110. The indicated probable error of the fitted curves is of the order of one percent for intermediate distances, increasing to from two to seven percent at the extremes of the distance range covered.

The curve for spherical charges is in agreement with the Kirkwood theory, and the results for cylinders having various length/diameter ratios indicate large dependence of pressure on charge shape for values of Z up to 50 at least.

I. INTRODUCTION

IT has long been recognized that pressure waves of finite amplitude in mediums like air propagate at greater than sound velocities, the velocity increasing with amplitude. For this reason, a continuous wave of finite amplitude changes its form with time, a pressure front becoming steeper until limited by the viscosity

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and heat conductivity of the medium. When this relatively stable form is reached, the pressure rise occurs within an extremely narrow region, and for most purposes may be considered a perfect discontinuity. Such a disturbance is called a shock wave. The conditions governing the propagation of such discontinuities were formulated by Rankine and Hugoniot, who gave expressions for the conservation of mass, momentum, and energy across the shock front.[1]

The spherical wave produced by the detonation in free air of an explosive charge is a wave of this type, having the characteristic form shown in Fig. 1 for a pressure jump of the order of 0.5 atmosphere.[2] The initial rapid rise to a peak pressure P is followed by a more gradual decay, a shallow suction phase, and a return to atmospheric pressure P_0, either gradual, or including a small secondary shock. The peak pressure decays as the wave propagates outward for two reasons: (1) because the energy must be distributed over a constantly increasing area, and (2) because there is a finite dissipation of energy in the transition through the shock front. Exact solution of the hydrodynamical equations, subject to initial conditions describing the release of energy in the explosive process and to the Rankine-Hugoniot conditions at the shock front, is not possible because of the non-linear nature of the equations of motion, and direct numerical integration is, in general, prohibitively laborious. In a recently published paper,[3] Kirkwood and Brinkley, by imposing a similarity restraint on the energy-time curve of the shock and using the second law of thermodynamics to determine, at an arbitrary distance, the distribution of the initial energy, have reduced the problem to two ordinary differential equations which are amenable to numerical integration. Their computations then provide a theoretical peak pressure *vs.* distance relation.

Experimental determination of the peak pressure of shock waves has also proved to be difficult. The most common method and that which was for a long time considered the most promising, has been the use of piezoelectric crystals, which produce a charge proportional to the applied pressure. The fundamental objection to such gauges, which has only recently become clear, results from the disturbance of the flow caused by the presence of the gauge itself. This effect makes the apparent pressure as read by the gauge less than the true pressure (i.e., the pressure that would exist in the absence of the gauge). The true pressure, moreover, is not a linear function of the gauge reading. Calibration of the gauges therefore becomes a difficult matter. These difficulties can be reduced considerably by suitable streamlining of the gauge and mount, but a complete solution requires a method of dynamic calibration exactly duplicating the conditions under which the gauge is used, which has not so far been achieved.

A technique for pressure measurements which is not subject to these objections has been suggested in the application of the relation between the peak pressure and the velocity of propagation of a shock. If gauges are used simply to time the passage of the shock over a known distance, the need for calibration is eliminated, and the local effect due to the presence of the gauges is unimportant so long as the over-all speed of propagation of the wave is not reduced. While the experiment requires careful planning to obtain satisfactory precision, the elimination of systematic errors should assure more reliable results than have been obtained previously.

The derivation of the pressure-velocity relation has been given in numerous places.[4] Using the

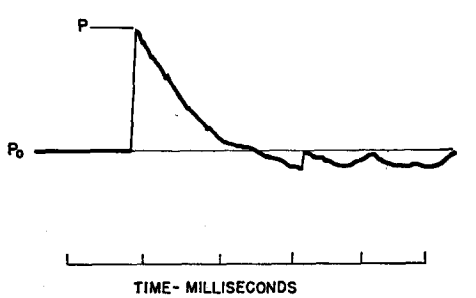

FIG. 1. Typical shock wave from a small charge. (8 feet from 0.5 lb. of TNT. $P-P_0 \sim 0.5$ atmosphere.)

[1] W. J. M. Rankine, Phil. Trans. **A160**, 277 (1870). H. Hugoniot, J. de l'ecole polyt. **57**, 3 (1887); **58**, 1 (1888).

[2] The form of very weak shock waves is somewhat different, and has been investigated by J. W. M. DuMond, E. R. Cohen, W. K. H. Panofsky, and E. Deeds, J. Acous. Soc. Am. **18**, 97 (1946).

[3] S. R. Brinkley, Jr. and J. G. Kirkwood, Phys. Rev. **71**, 606 (1947).

[4] G. I. Taylor and J. W. Maccoll, "Dynamics of Compressible Fluids," from *Aerodynamic Theory*, ed. by W. F. Durand (Julius Springer Verlag, Berlin, 1935) Vol. III, Div. H; H. W. Liepmann and E. E. Puckett, *Aerodynamics*

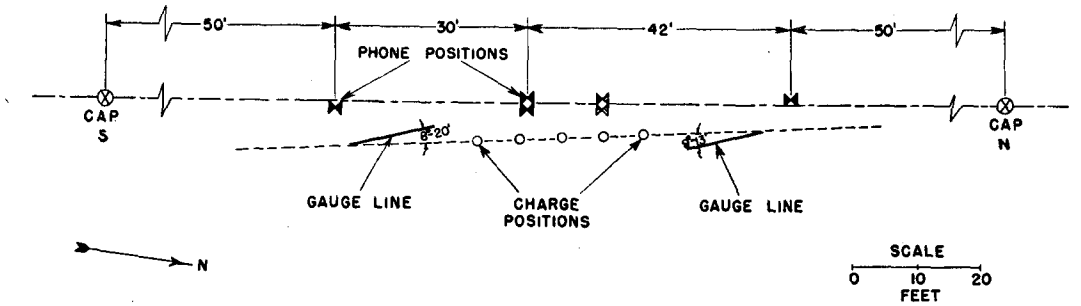

FIG. 2. Plan of experimental set-up. Ten gauges were mounted uniformly, 18 inches apart, along each gauge line.

Rankine-Hugoniot relations and calculating the internal energy of the gas from the ideal gas law, we obtain

$$(P-P_0)/P_0=[2\gamma/(\gamma-1)][(V^2/a^2)-1], \quad (1)$$

where P is the absolute pressure immediately behind the shock front, P_0 is the atmospheric pressure ahead of the shock, V is the velocity of the shock relative to the medium ahead, a is the velocity of sound in this medium, and γ is the ratio of specific heats C_P/C_V. At high pressures the ideal gas approximation with constant γ in the energy equation is not correct. Comparison of Eq. (1), using $\gamma=1.40$ for air, with more exact calculations[5] indicates, however, that for an excess pressure $P-P_0$ of 100 lb./in.^{2} the error in $P-P_0$ is less than 0.7 percent and that this error diminishes rapidly for lower pressures.

Examination of Eq. (1) shows that the ratio of the percent error in excess pressure $P-P_0$ to the percent uncertainty in the velocity ratio V/a is never less than two, and is very large for pressures only slightly greater than P_0. In the pressure region of interest in this investigation ($\sim$0.1 to 10 atmos. excess) the error ratio is about ten in order of magnitude. Thus in order to determine the excess pressure within 1 percent, velocity measurements must be good to about 0.1 percent. This was taken as the design criterion, and the experiment was planned so that this precision could be expected in all the measured quantities involved in determining the velocity ratio.

II. EXPERIMENTAL PROCEDURE

To measure shock wave velocity, quartz piezoelectric gauges were chosen as being best suited to the requirements of the problem. These consist of an x-cut quartz disk, 0.9 inch in diameter and 0.1 inch thick, mounted in a sealed brass housing with a thin dural face plate which communicates the external pressure to the x face of the crystal. Used "face-on" (the blast incident normally on the sensitive face) these gauges provide an extremely fast response to pressure changes, and the flat face furnishes a definite point from which to measure the baseline distance. Ten such gauges were mounted rigidly at intervals of 18 inches and connected in parallel on a single channel of a multichannel cathode-ray oscilloscope unit. This was part of a mobile unit developed especially for blast recording, and provided amplifiers with good high frequency response and exceptionally well focused traces so that the time of arrival of a shock wave at a gauge could be determined within about a micro-second.

Two lines of ten gauges each were set up on opposite sides of the charge, with a distance of 40 feet between the first gauge of each line, as shown in Fig. 2. With a spacing of 18 inches between gauges, a total distance range of about 7 feet to 47 feet could be covered by moving the charge position back and forth between the two gauge lines, 18 pressure points being obtained from the two gauge lines on each shot. Charge positions were located at the point midway between the gauge lines, and symmetrically on each side of this point. In this way the two independent sets of gauges could be checked against each other directly (charge in center position) and in effect interchanged in covering distances close to and far from the charge by taking shots at corresponding positions on opposite sides of the center. Excellent agreement was observed between the results of the two groups of gauges.

of a Compressible Fluid (John Wiley and Sons, Inc., New York, 1947).

[5] J. G. Kirkwood, S. R. Brinkley, Jr. and J. M. Richardson, unpublished work.

To reduce the risk of the nearer gauges blanketing those behind, each gauge line was rotated at an angle of about 9° from the line joining the farthest gauge of each line. A test showed that with the gauges staggered to this extent there was no observable attenuation of the shock wave due to its passage along the gauge line. Careful measurements of the angle and distance of each gauge line with respect to each charge position were made, as well as of the intergauge distances along each gauge line, so that the differences in the radial distances of successive gauges from the charge could be calculated. These represent the true baselines over which the spherical shocks were timed. When the charge position was changed each gauge was rotated about a vertical line through the center of its sensitive face so as to remain normal to the radial line from the charge.

Short pulses of about five microsecond duration were superimposed at a rate of 10,000 per second on each recording channel to facilitate timing the intervals between the arrival of the shock at successive gauges. These pulses were produced by a crystal controlled oscillator checked against the standard carrier frequency of station WWV.

The charges used were either spherical or cylindrical in shape, weighing between $\frac{1}{2}$ and 8 pounds. The cylinders were hung with their axes vertical, so that all the gauges were in the equatorial plane of the charge. In measuring "free air" peak pressure, the charge and gauges must be sufficiently far above the ground to eliminate interference with the incident wave arising from reflection phenomena at the ground. In the present experiments the gauges and charges were located in a horizontal plane 10 feet above the ground, a height sufficient to guarantee that the effect of the ground could not be felt until the shock wave had passed the farthest gauge.

The velocity of sound in the medium must also be known in applying the pressure-velocity relation. Because of the difficulty of controlling or measuring random local variations in the temperature, humidity, local velocity, etc., of the outdoor atmosphere, which affect the velocity of both sound and shock waves, the velocity of sound was measured directly and as nearly as possible at the same time and place as were the shock waves themselves. Ordinary telephone receivers were set up as indicated in Fig. 2 to provide two independent baselines of 30 and 42 feet over which the velocity of the wave produced by the detonation of a blasting cap could be measured. The caps were located at a sufficient distance for the velocity of the pressure wave to have decayed to sound velocity within the limits of observation. Two caps were detonated in close succession at positions N and S (Fig. 2) to provide determinations of sound

TABLE I. Charges used in experimental program.[a]

Designation	Composition	Average[b] weight (pounds)	Shape	Height[c] (inches)	Diameter[c] or width (inches)	Height/diameter ratio	Number of shots	Number of pressure observations
Q	Pressed TNT	0.5	Rectangular block	3.50	1.75	—	17	117
A	Cast Pentolite[d]	8.00[f]	Cylinder	8.75	4.50	1.94	5	88
B	Cast TNT[e]	4.10[g]	Cylinder	7	3.62	1.93	8	144
C	Cast Pentolite[d]	3.75	Sphere	—	5	—	8	142
D	Cast TNT[e]	2.89[h]	Cylinder	5	3.62	1.38	5	90
E	Cast TNT[e]	2.89[h]	Cylinder	12	2.31	5.20	5	90
F	Cast TNT[e]	1.45[i]	Cylinder	2.5	3.62	0.69	1	18
G	Cast TNT[e]	1.45[i]	Cylinder	6	2.31	2.60	3	52
H	Cast TNT[e]	1.45[i]	Cylinder	8	2.00	4.0	3	45
K	Cast TNT[e]	1.45[i]	Cylinder	12	1.62	7.4	1	18

[a] The spherical pentolite charges were prepared by the Woods Hole group of Division 2, NDRC. All cylindrical charges were supplied by the Picatinny Arsenal.
[b] The weights of individual charges varied by as much as ±5 percent in some cases from the average values listed. Accurately measured weights were used in scaling to reduce results to a common basis.
[c] Dimensions given are nominal values. Variations of the order of ±5 percent occur in some cases.
[d] 50 percent TNT and 50 percent PETN, by weight. Specific gravity ~1.60.
[e] Specific gravity of cast TNT ~1.55.
[f] Includes 0.031-lb. tetryl pellet booster.
[g] Includes 0.096-lb. tetryl pellet booster.
[h] Includes 0.067-lb. tetryl pellet booster.
[i] Includes 0.030-lb. tetryl pellet booster.

velocity in opposite directions, from which the velocity of sound relative to still air and the component of wind velocity in the direction parallel to the gauge lines could be determined. The effect of cross wind velocity (although it was not measured) is entirely negligible since it is a second-order effect, and, by good fortune, the prevailing wind direction during the period of shooting was very nearly parallel to the gauge lines, so that the cross-wind component was in any case only a fraction of the parallel wind component which was measured. The two caps and the main charge were set off automatically at intervals of about 0.1 second to eliminate as far as possible the danger of fluctuations in the atmosphere between the time of the sound measurements and the shock measurements.

The proper wind velocity correction was added to the shock velocity measured over each interval of the gauge line, and the corrected shock

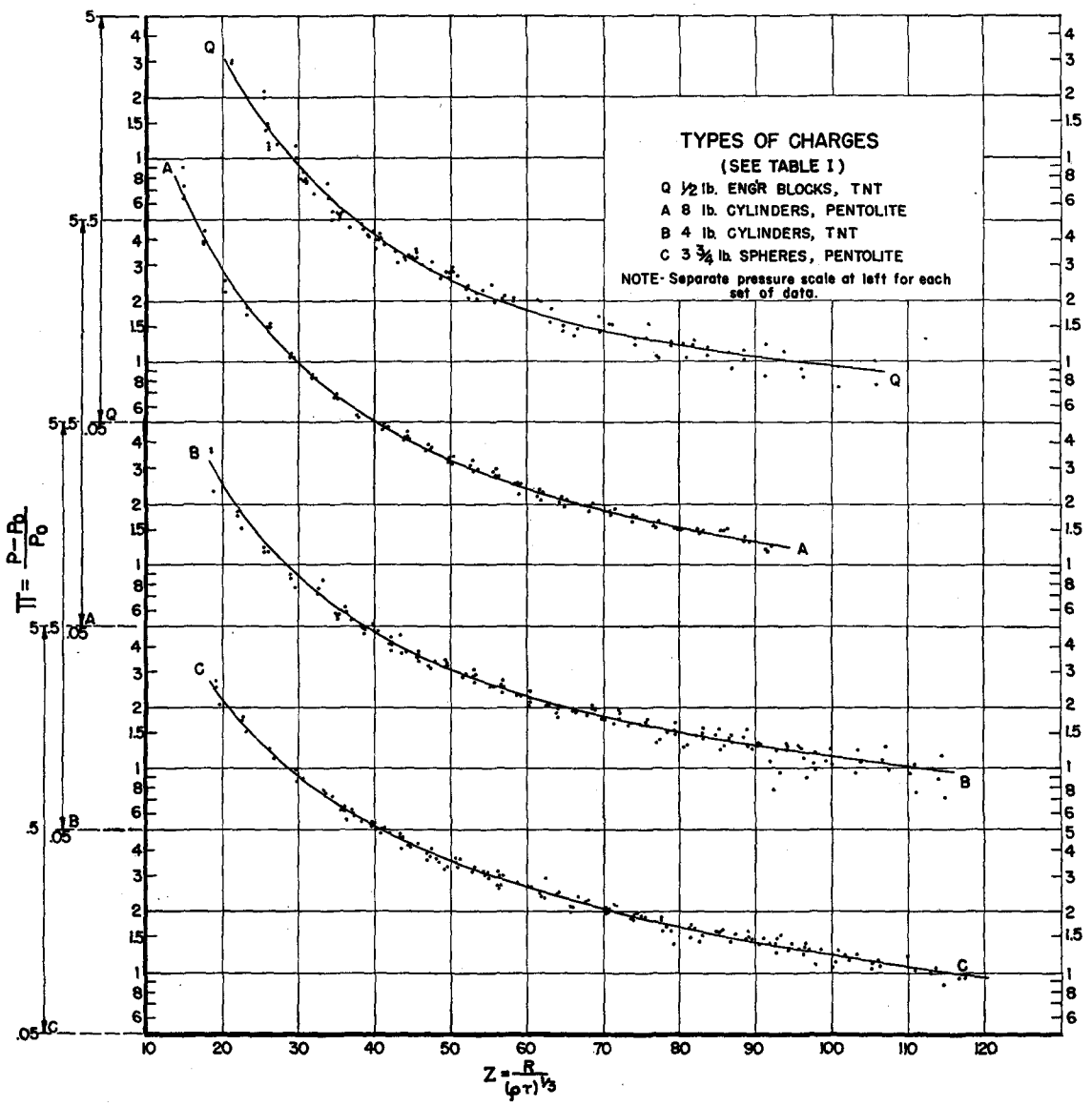

FIG. 3. Experimental pressure *vs.* distance results for four types of charges. Excess shock wave pressure in non-dimensional units, $\Pi=(P-P_0)/P_0$, is plotted against the non-dimensional distance $Z=R/(\rho\tau)^{\frac{1}{3}}$. For specific gravity $\rho=1.60$, $Z=1.06R/D$, where D is the diameter of a sphere having the same volume as the charge. Thus Z represents approximately the distance in charge diameters, for the explosives used. (For TNT, $\rho=1.55$; for pentolite, $\rho=1.60$.)

TABLE II. Fitted curves giving best values of experimental pressure measurements on four series of charges.

Type of charge	Distance range covered by experimental data $[Z=R/(\rho\tau)^{\frac{1}{3}}]$	Fitted curve for pressure $\Pi=(P-P_0)/P_0$	Standard deviation[a] of Π (expressed in percent) σ_1	σ_{min}	σ_2
TNT Eng'r Blocks (Q)	$21<Z<107$	$\Pi_Q=13.50/Z-769.9/Z^2+36280/Z^3$	±3.9%	±2.1%	±4.9%
Pentolite Cylinders (A)	$14.7<Z<91$	$\Pi_A=10.49/Z-135.5/Z^2+21070/Z^3$	7	0.9	1.4
TNT Cylinders (B)	$18.6<Z<111$	$\Pi_B=11.34/Z-185.9/Z^2+19210/Z^3$	6.5	0.8	2.2
Pentolite Spheres (C)	$18.6<Z<119$	$\Pi_C=8.63/Z+295.1/Z^2+7823/Z^3$	2.1	0.6	1.4

[a] σ_1 is the standard deviation at the smallest Z indicated in column 2 and σ_2, at the largest Z. σ_{min} represents the minimum value of standard deviation reached usually at about the midpoint of the Z range. The curve for σ is generally flat in the neighborhood of σ_{min}, and the standard deviation is roughly $<1.5\sigma_{min}$ in the region $40<Z<90$.

velocity V was inserted in Eq. (1), along with the measured sound velocity a, and the corresponding pressure calculated. Since the shock velocity changes appreciably with distance, the pressure calculated from the measured average velocity over the interval corresponds to the instantaneous pressure at a point within the interval determined by the relation

$$V(R_V)=(1/R_m q)\int_{R_{m-}}^{R_{m+}} V(R)dR, \qquad (2)$$

where R_V is the distance from the charge at which the instantaneous pressure equals the pressure computed from the average velocity, R_m is the distance from the charge to the midpoint of the interval over which the velocity is measured, q is the ratio of the length of the interval to R_m, $R_{m+}=R_m(1+q/2)$, $R_{m-}=R_m(1-q/2)$, and V is the function expressing shock velocity in terms of distance from the charge.

An approximate expression for R_V may be obtained by assuming that the pressure over a short interval may be represented by an expression of the form $P-P_0=AR^{-n}$, where A and n are constants, and expanding the resulting expression for $V(R)$ obtained by substituting this in Eq. (1). The expression valid for low pressures was first given by Carr, Schwarzschild, and Weiss, working at Aberdeen, and has the form

$$R_V=R_m\{1-(1+n/2)q^2/24\}. \qquad (3)$$

A slightly different expression is obtained when an expansion valid at high pressures is used. A more detailed derivation of Eq. (3) shows that it actually holds for moderately high pressures, and that in the most unfavorable case in the experiments reported here the error in R_V is only 0.18 percent.

III. RESULTS

The physical characteristics of the types of charges investigated are given in Table I. In order to compare results from charges of different weights, the data have been reduced to a common basis by scaling the distances according to the principle of similitude. For similar charges of equal density, equal pressures are expected at distances proportional to the linear dimensions of the charges, and hence to the cube root of the charge weight. This scaling law has been shown experimentally to be valid over a large range of charge weights. The scaled distances are given in terms of the non-dimensional variable $Z=R/(\rho\tau)^{\frac{1}{3}}$, where R is the actual distance from the charge, having a volume τ and specific gravity ρ relative to water.[6] The pressure is expressed non-dimensionally in units $(P-P_0)/P_0$, denoted hereafter by Π.

The charges designated A, B, and C in Table I formed the major part of the experimental program, and the complete sets of pressure-distance points for these three series and for the pre-

[6] It has been customary in the past to represent the scaled distance by the dimensional variable $z=R/w^{\frac{1}{3}}$ where R is the distance in feet, and w is the charge weight in pounds. The quantity z thus has dimensions ft./(lb.)$^{\frac{1}{3}}$. The relation between z and the non-dimensional variable $Z=R/(\rho\tau)^{\frac{1}{3}}$, which we have introduced, is obtained by replacing w by $62.43\rho\tau$. It follows that $Z=(62.43)^{\frac{1}{3}}z=3.967z$. Thus a distance in Z units when divided by 4 represents roughly the distance in feet from a one pound charge.

The convenient relationship between the Z units and charge diameters may be seen by substituting $V=\pi D^3/6$ where D is charge diameter. We then have $Z=(\pi\rho/6)^{-\frac{1}{3}}\times(R/D)$. Thus Z represents exactly the number of charge diameters (R/D) when $(\pi\rho/6)^{-\frac{1}{3}}=1$; i.e., for $\rho=6/\pi=1.910$. Because of the cube root involved, Z also represents approximately the distance in charge diameters for a considerable range of values of ρ. For $\rho=1.50$, for example, $(\pi\rho/6)^{-\frac{1}{3}}=1.084$, and Z differs from the number of charge diameters by 8.4 percent.

The "density of loading," as the term is customarily used, is the same as the specific gravity ρ used here. Hence R will be in feet if τ is in cu. ft., in cm if τ is in cc, etc.

liminary series (Q) using U. S. Corps of Engineers Demolition Blocks of pressed TNT are shown as small dots in Fig. 3, A, B, C, and Q, respectively.

The curves shown represent least square fits of an expression of the form $\Pi = AZ^{-1} + BZ^{-2} + CZ^{-3}$, introduced by Reynolds,[7] which has been found to give a satisfactory representation of the data over the range of Z involved. Asymptotic solutions of the propagation equations show a decay in pressure proportional at large distances to Z^{-1} except for a slowly varying logarithmic factor, and experimental measurements in general point to Z^{-3} as a possible limiting value for the dependency at small distances. The Z^{-2} term provides a third parameter necessary for fitting the data successfully. Since the pressure in the explosive is known to be finite, an expression of this form cannot hold for very small Z. For this reason the fitted curves should be extrapolated only with great caution beyond the range of the experimental data. The equations of the fitted curves for the first four groups of charges in Table I are given in Table II. The limiting values of Z indicate the range covered by the experimental data. The quantity σ is an estimate of the reliability of the fitted curve, expressed as a standard deviation; that is, if the experiment were repeated, in $\frac{2}{3}$ of the cases the value of Π given by the fitted curve would lie within $\pm\sigma$ of the "true" curve which would be obtained from a very large number of observations in the same kind of experiment. The values of σ given in Table II are expressed as a percentage of the value of Π as given by the fitted curve, which represents the best value of the pressure as indicated by the experimental data. The standard deviation of an individual point is ~ 5 times the σ for the curve. The statistical treatment by which these σ's were calculated took into account the non-independence (in the statistical sense) of data from adjacent gauge line intervals.[8]

The relationship of the various experimental determinations to each other and to the theory are shown in Fig. 4. To emphasize the comparison of the various curves, the pressure curve for pentolite spheres (Π_c in Table II) has been chosen arbitrarily as standard, and the difference of each of the other curves from it, expressed as a percentage of excess pressure, is plotted as ordinate. The ordinate is thus the quantity $100(\Pi - \Pi_c)/\Pi_c$ (equivalent to $100(P - P_c)/(P_c - P_0)$). The abscissa is the non-dimensional scaled distance Z. Thus the curve for pentolite spheres is represented by the horizontal axis.

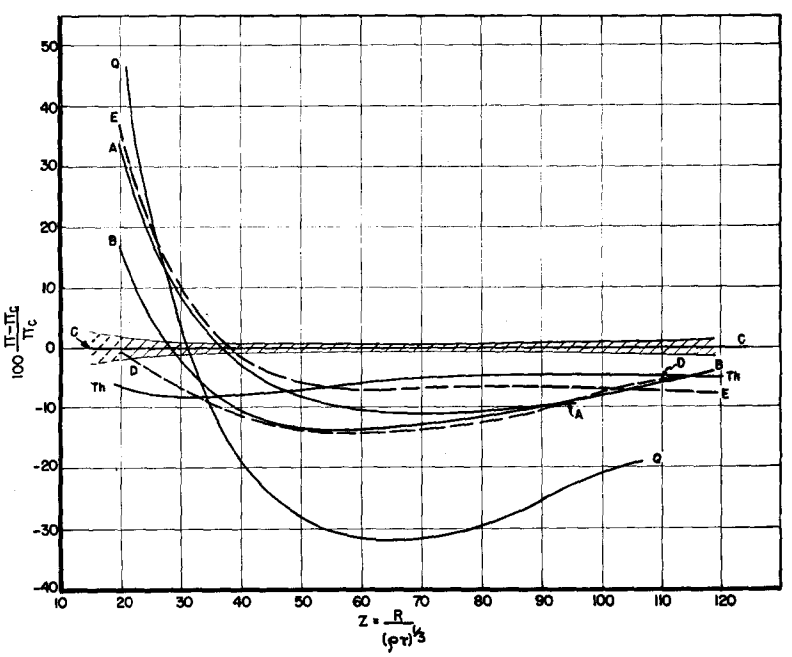

FIG. 4. Comparison of results. Each curve represents the percent deviation of excess pressure, $100(\Pi - \Pi_C)/\Pi_C$, from the value for pentolite spheres Π_C, plotted as a function of the non-dimensional distance Z. The experimental curves are designated by letters as in Table I. Curve *Th* represents Kirkwood and Brinkley's theoretical prediction for TNT spheres.

[7] G. T. Reynolds, "Small charge air blast experiments," OSRD Report No. 1518 (1943).
[8] We are indebted to Dr. H. Sheffe for developing a form of statistical treatment appropriate to this problem.

(The choice of this curve as standard is only for convenience in making the greatest number of meaningful comparisons and, of course, implies no fundamental distinction in the reliability of this curve relative to the others.) The shaded region indicates the standard deviation band $\pm\sigma$ (expressed in percent). Bands of similar width apply to the other experimental curves. The various curves are identified according to the designations used in Table I.

Curve *Th* represents the theoretical curve for spherical charges of TNT computed by Kirkwood and Brinkley on the basis of their recently reported theory.[3] A curve for pentolite, which could be compared directly with the experimental results, is not available, but experimental work at Woods Hole indicates that 5 percent probably represents the superiority in pressure level of pentolite over TNT, averaged over the whole distance range involved. Thus the approximate constancy of the difference between curve *Th* and curve *C*, with an average value of from five to seven percent, represents good agreement between the theory and the experimental data.

The effect of charge shape may be seen directly from curves *C* and *A* which represent pentolite spheres and cylinders, respectively. Curve *B* is for TNT cylinders having the same length/diameter ratio as the pentolite cylinders. Its approximate similarity to curve *A* is a further confirmation that the observed difference between cylinders and spheres is real, but the two curves raise some question whether (at least in the case of cylindrical charges) the ratio of the pressures from TNT and pentolite is a constant, independent of distance. The shape of curves *A* and *B* may be explained qualitatively by the observation that the shock wave from the sides of a cylindrical charge should at first be more nearly cylindrical than spherical, and thus show a slower decay and relatively higher pressure at small distances from the charge. At some point, the rarefaction propagating in from the ends would begin to take effect, and the wave could then be expected to decay more rapidly than the corresponding spherical wave. It is somewhat surprising that this effect is apparently so pronounced at distances greater than 20 charge diameters, but in view of the indicated precision of the experimental determinations, the observed effect must be considered real. The explanation suggested is of course only qualitative.

The remaining types of charges in Table I, consisting of smaller cylinders of TNT having a larger range of length/diameter ratios, were not used in sufficient numbers to make it advisable to treat the data statistically, but curves could be fitted by eye with reasonable assurance. Because of the smaller size of the charges and the smaller number of observations, the results are less reliable, but are almost certainly good to better than 10 percent, except at the extremes of the distance range. Curves *D* and *E* are for 2.9-lb. TNT cylinders with length/diameter ratios of 1.38 and 5.20, respectively. The former is somewhat less than the ratio for the charges of curve *B*, the latter considerably larger. The relative values of the three curves at small distances are consistent with the argument suggested above, and the three curves approach agreement at large distances, which is to be expected. The data from 1.45-lb. cylindrical TNT charges with four different length/diameter ratios varying from 0.69 to 7.4 also showed similar behavior, but the number of shots was so small (only 2 shots, providing 18 pressure points each, with no overlap in distance, in the case of the smallest and the largest L/D ratios) that an accurate comparison is of little significance, and the curves have not been included in Fig. 4.

An extreme example of the effect of charge shape is seen in curve *Q*, representing measurements made with standard Corps of Engineers Demolition Blocks of pressed TNT, whose measurements are given in Table I. These blocks have a square cross section, with slightly rounded corners, and measurements were made in the equatorial plane in a direction normal to a flat face. Thus the effect of both corners and ends may be expected, and a large divergence from the pressure-distance law for spherical charges is observed, considerably larger than in the case of even the most elongated cylinders investigated.

There have been published, as far as the authors are aware, no experimental results with which the data in this paper can be compared. They are aware, however, of the existence of other results to which reference cannot be made at this time because they have not been published.

IV. CONCLUSION

The velocity method of measuring shock wave pressures which has been developed appears to offer a means of determining absolute peak pressure which is (1) free from the systematic error inherent in direct gauge measurements due to the physical presence of the gauge in the flow, and (2) capable of precision of better than 1 percent in the excess pressure when a sufficiently large number of measurements is made. A fundamental limitation appears to be the nonuniformity of the atmosphere under even the best field conditions. On the basis of such measurements, Kirkwood's theory has been checked experimentally, and a difference in the decay of pressure with distance arising from the shape of the charge has been observed which persists at relatively large distances from the charge.

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Chromatic Aberration and Resolving Power in Electron Microscopy

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Contrast difficulties in the observation of fine detail in thin specimens may be overcome by employing dark-field observation. This introduces the added complication, however, that the imaging electrons are now inhomogeneous, having experienced a most probable energy loss of the order of 20 volts. The present calculations of the effect of chromatic aberration and diffraction show that with an objective aperture smaller than the "optimum aperture," and centered optics, such inhomogeneities do not appreciably affect the resolution, even in the extreme case that the electrons are uniformly distributed in energy.

THE recognition of fine detail attainable in electron micrographs depends on the ultimate resolving power of the instrument, as determined by lens aberrations[1] and diffraction effects, on the one hand, and by the image contrast under the given operating conditions on the other. The condition generally regarded as favorable for attaining the highest resolutions—namely, bright-field observation of very thin objects with a physical objective aperture at least equal to the "optimum aperture"—inherently leads, unfortunately, to relatively low contrasts. Apart from contrast limitations, the resolution is here limited by spherical aberration and diffraction, leading, according to Glaser's wave-mechanical theory,[2] to the expression

$$d_{\min}=0.56\lambda(Cf/\lambda)^{\frac{1}{4}} \tag{1}$$

for an "optimum aperture"

$$\alpha_{\text{opt}}=1.13(Cf/\lambda)^{-\frac{1}{4}}. \tag{2}$$

Here $d_{\min}$ is the least resolvable separation of two object points, C the dimensionless spherical aberration constant of the objective, f its focal length, and λ the de Broglie wave-length of the imaging electrons. Equation (1) represents a lower limit to the resolution obtained with the

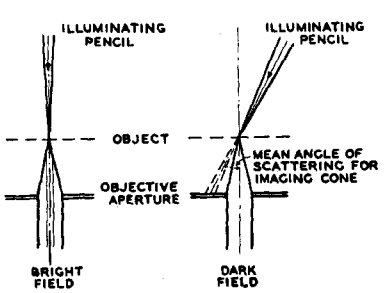

Fig. 1. Arrangement of illumination for bright-field and dark-field observation.

[1] The effects of field asymmetries as well as voltage fluctuations (effective because of the chromatic aberrations of the lenses) are included herein.

[2] W. Glaser, "Image formation and resolving power of the electron microscope from the standpoint of wave mechanics," Zeits. f. Physik 121, 647 (1943).